The Low-Budget Advantage: A Data-Driven Case Against Hollywood's Blockbuster Bias

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1. INTRODUCTION

Since the birth of the blockbuster with *Jaws*, Hollywood studios have been chasing after the next big, expensive hit. Over the years, the film industry has seen massive successes and massive losses through the blockbuster. But in the current media landscape, where Hollywood is fighting to stay relevant and entice audiences back to the theater, are big-budgeted tentpole films still the way forward, or have mid-budgeted and low-budgeted films become a safer bet for studios? Drawing on both box office data (2000-2024) as well as my ten years in the film industry, I will identify which category of film is the smartest investment within the current landscape.

KEY STRATEGIC QUESTIONS

- Which budget class shows the highest profits and ROI?
- Does IP have a significant effect on each budget class?
- Which types of film are the safest investments?

2. DATA SOURCING AND METHODOLOGY

Data Sourcing

I sourced a dataset on Kaggle of box office records. The dataset was comprehensive, containing data values for release date, budget, domestic gross, international gross, inflation adjustment, genre, and franchise. However, the dataset did contain some noticeable inconsistencies for titles released before 1969 and in 2025. This stems from inconsistent records for pre-1969 titles and still evolving records for 2025 releases. Given that I wanted to look at the modern landscape, I selected the records for films released between 2000 and 2024, thus keeping the analysis focused while at the same time sidestepping the issue of those inconsistent records.

Data Cleaning

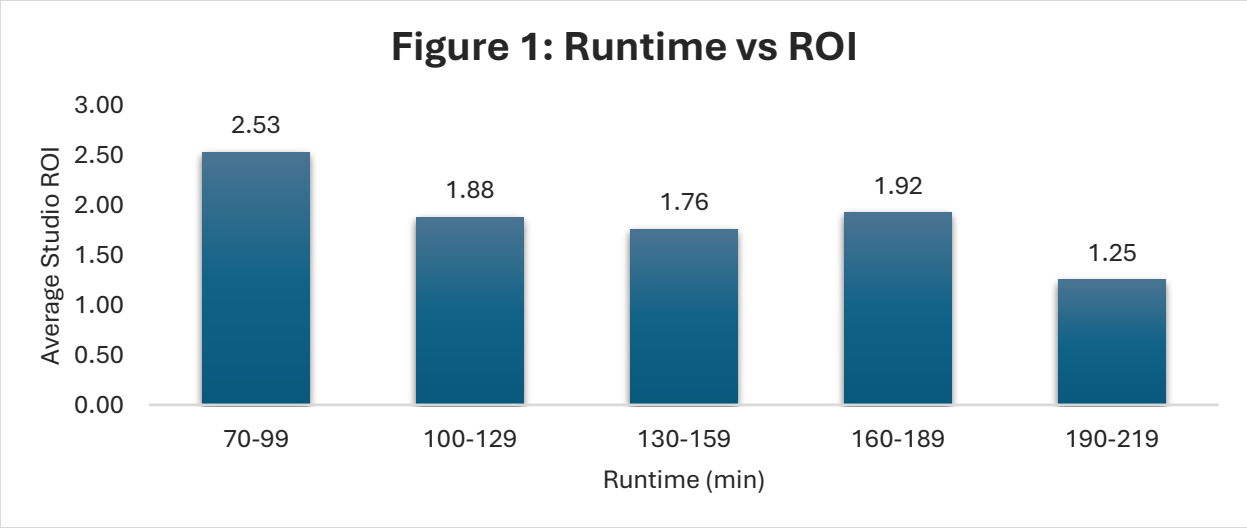
I performed data cleaning using SQLite, removing duplicates and standardizing formats and number values. NC-17 or Not Rated entries often included additional descriptive text, so those string values were standardized to just contain the rating. All entries with release dates between 2000-01-01 and 2024-12-31 were selected. However, some titles released from the 1920's and 1910's were mistakenly listed as 2020's and 2010's respectively. Fortunately, there was an additional column of string data that contained the date written out correctly, so I used that column to filter out those titles.

I then executed multiple calculations, including inflation adjustments, estimated profit, and studio ROI. The estimated profit was calculated via the industry standard benchmark of subtracting 2.5 times the production budget from the total gross, to account for marketing costs as well as the theater-distributor split. The studio ROI was calculated as 50% of total worldwide gross (again to account for the theater-distributor split), divided by the budget. Lastly, titles with a worldwide gross under $1M were filtered out for statistical integrity.

For a more detailed breakdown of my SQL cleaning process, refer to the appendix at the end of the document.

Runtime Null Values

Upon importing my cleaned data into Excel, I found that 198 of the total 3,979 titles contained null runtime values. While I was not planning on structuring much of my analysis around runtime, I wanted to make sure I wasn't neglecting a potentially crucial factor with incomplete data. Below is a histogram of runtime against ROI.

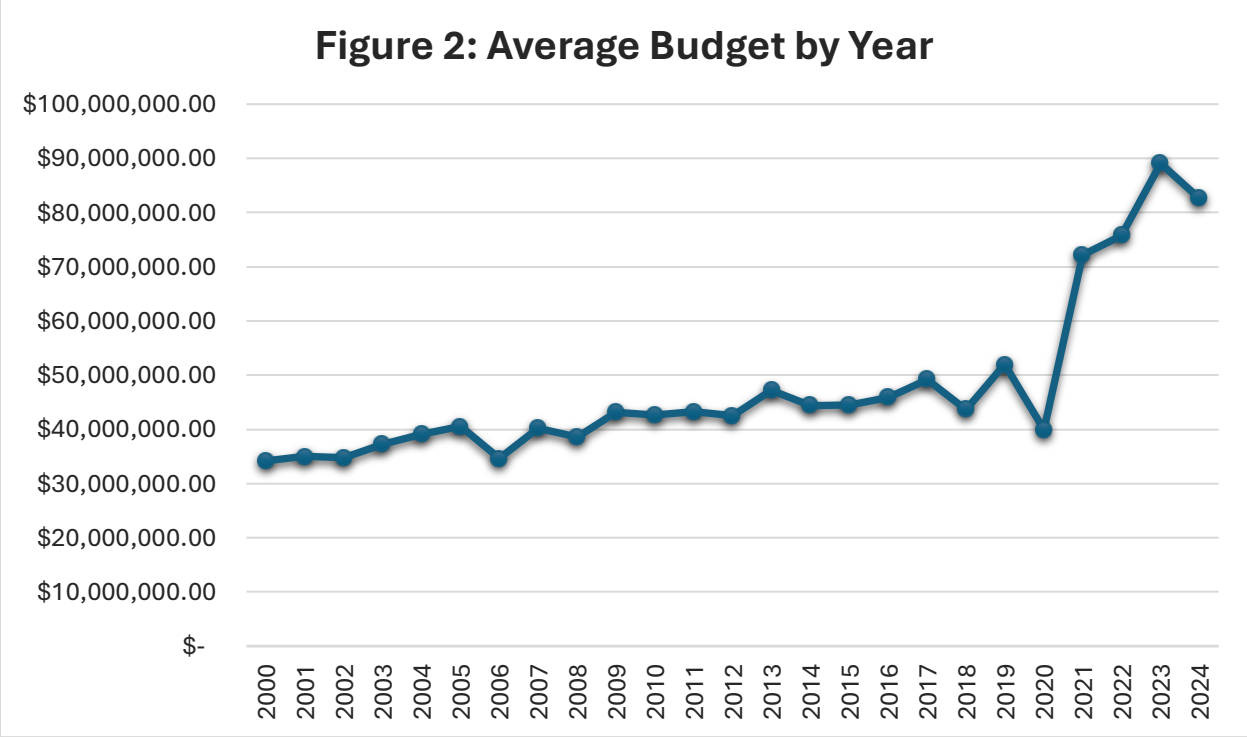


Since runtime shows no correlation with ROI, null runtime values were retained and runtime excluded from further analysis.

3. OVERVIEW: 21ST CENTURY FILM LANDSCAPE

Average Budget by Year

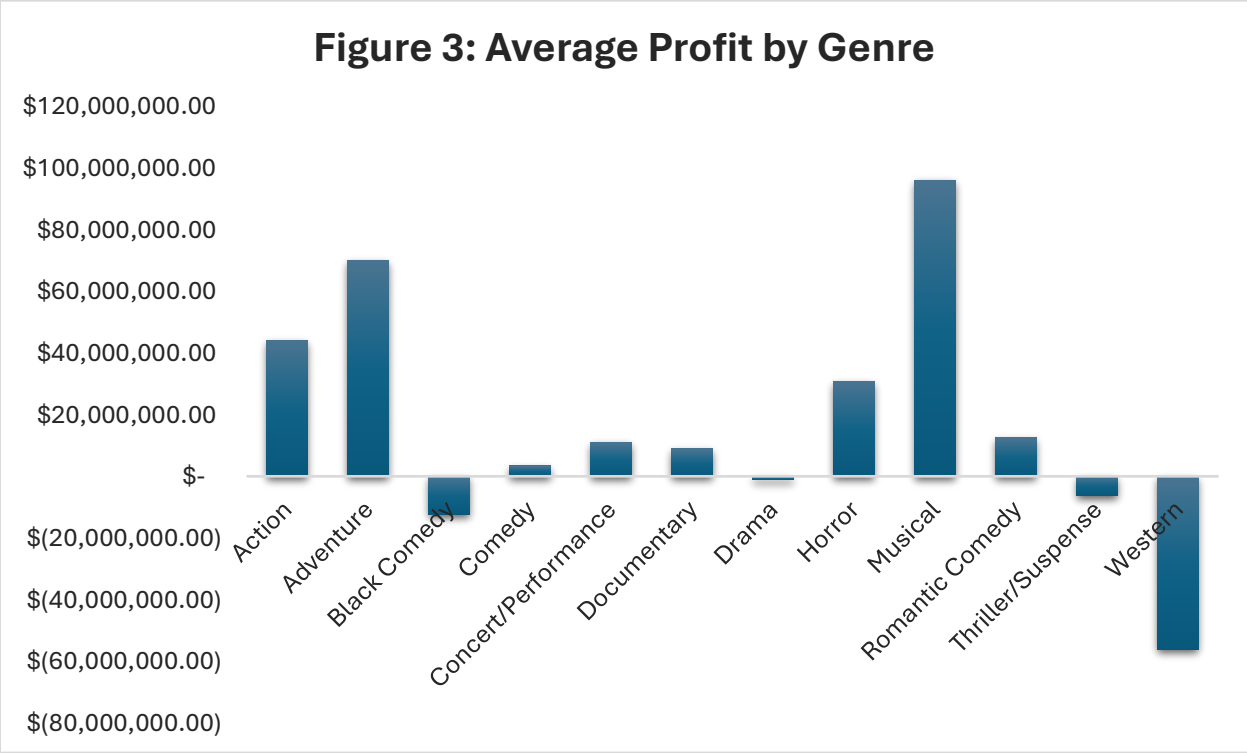
With our data fully cleaned, I generated three different exploratory Pivot Charts to construct an accurate overview of the industry's current state and trajectory. I began by creating a line chart of the average film budget per year.



There appears to be a general creep in budget over time, as to be expected; however, following the pandemic there was a sharp increase in average budget. At least some, if not most, of this spike can be attributed to films that were in production during the height of the COVID-19 outbreak. These films faced additional costs from temporary shutdowns to prevent infection spread, staffing additional safety crew members, and longer shoot schedules to accommodate the new safety precautions. Yet post-vaccine rollout, there appears to be only a slight dip in average budget in 2024; it remains unclear whether that is the start of budgets equalizing back down to a healthier number, or only a slight dip that could signal a worrying, permanent inflation.

Average Profit by Genre

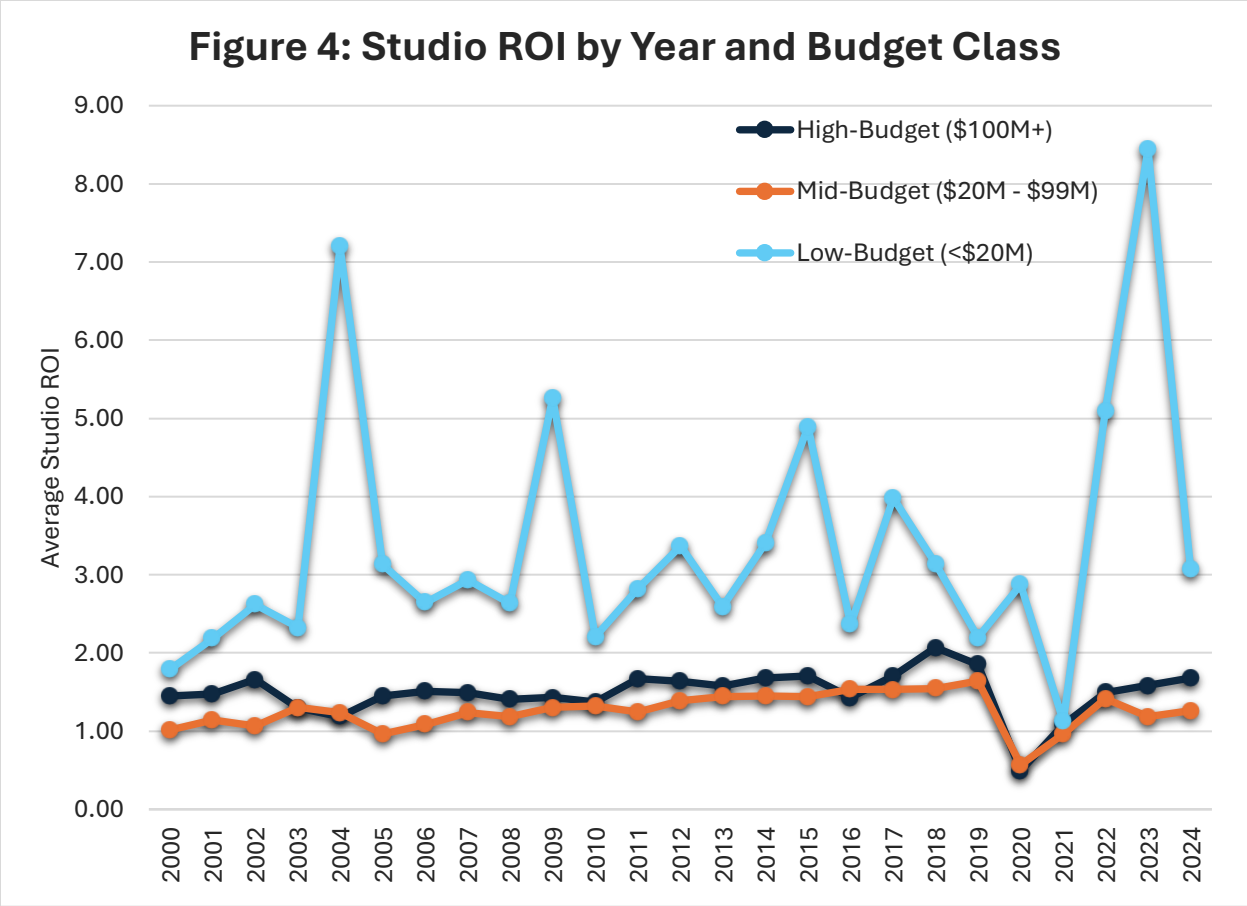
Subsequently, I produce a histogram visualizing the average estimated profit by genre.



Action, Adventure, and Horror all predictably show healthy levels of average profit. Surprisingly, Musical shows the highest average profit. Upon reflection, this high metric is most likely driven up by the animated musicals, such as *Moana* (2016) and live-action remake musicals, such as *The Lion King* (2019), from Disney, rather than original live-action musicals, of which there are relatively few. Western is by far the least profitable genre, showing significant losses on average. This aligns with current industry perceptions of the genre, given both how expensive and how out-of-fashion films in that genre are perceived to be.

ROI by Year and Budget Class

Lastly, I create a Pivot Chart mapping average studio ROI by year and budget class.

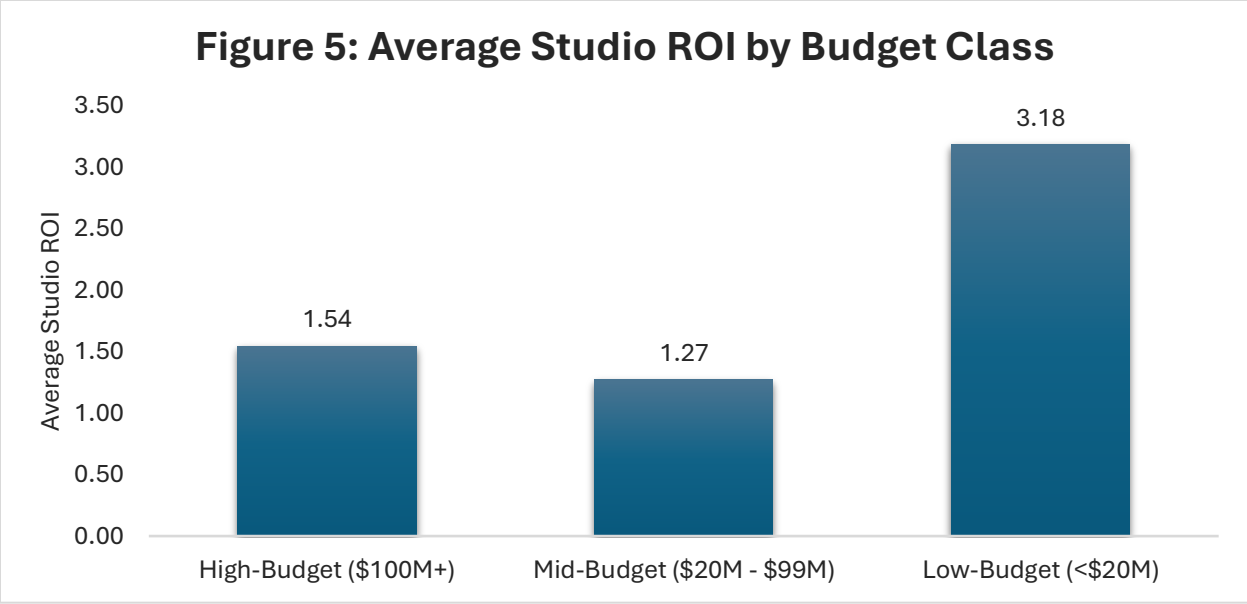


Average ROI on low budget films seems to be the most unstable, with numerous ups and downs. These spikes are presumably driven by individual crossover hits, as low-budget films lose relatively little when failures but stand to gain quite a lot when successes. However, what's most notable is that despite the ups and downs, the average ROI for low-budget films is always higher than the average ROI for high-budget and mid-budget films. Also of note is that, aside from the height of the pandemic in 2020, which saw a major decrease across all three budget categories, high-budget films tend to have a marginally higher average ROI than mid-budget films. These results warrant a closer examination and pointed analysis centered around budget class.

4. HIGH AND LOW: THE BUDGET CONUNDRUM

ROI by Budget Class

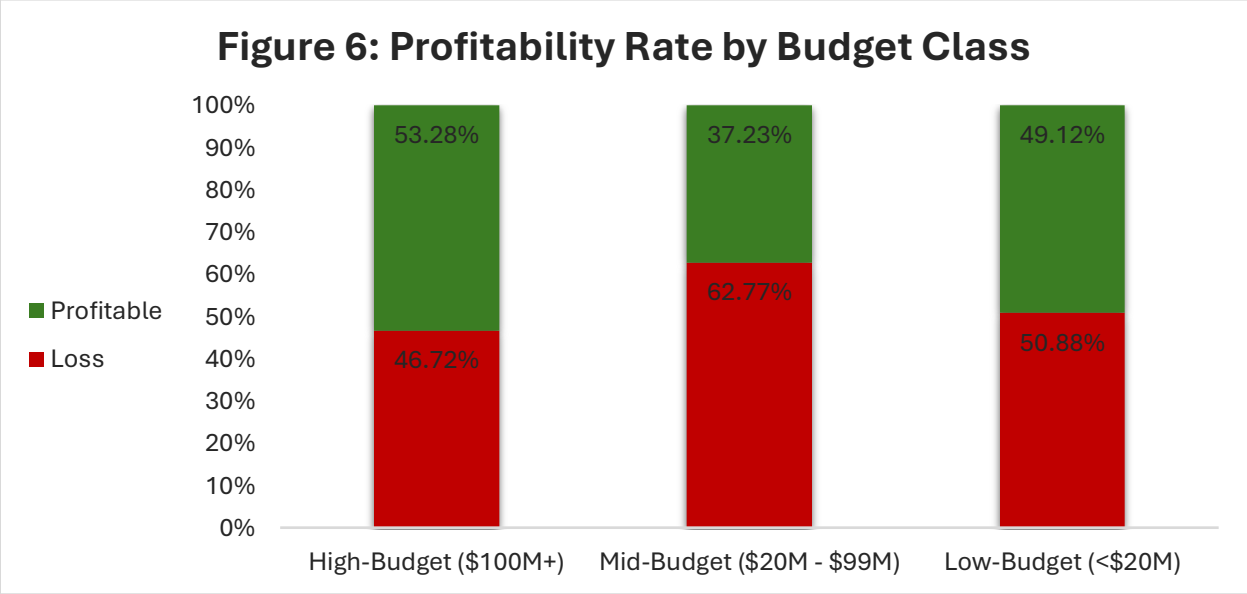
To compare the financial success of the budget classes, I generate a bar chart comparing the calculated ROI for each of the three classes. As noted earlier, ROI was calculated as 50% of total worldwide gross divided by budget.



Low-budget films are the clear leader in ROI, with both high-budget and mid-budget films considerably below it. This finding aligns with the logic that low-budget films are low-risk investments, in that they are not huge money losers when they underperform, due to their modest budgets, but still have the potential to become huge successes. However, the results are noticeably different when comparing profitability rather than ROI.

Profitability Rate by Budget Class

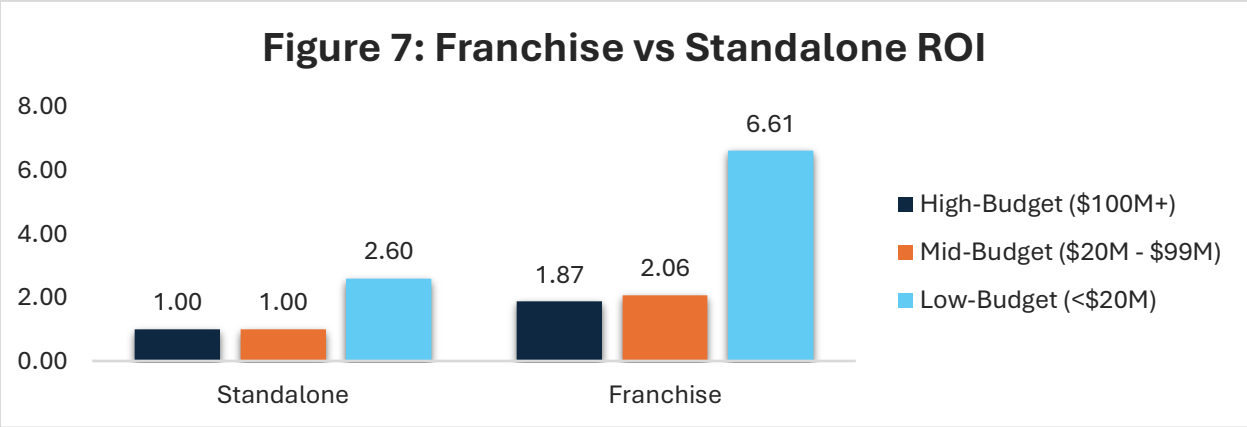
Next, I created a stacked bar chart to compare the profitability of the three budget categories. As mentioned earlier, profitability was estimated using the industry benchmark of subtracting 2.5 times the budget from the total worldwide gross.



Here, high-budget films are the leader, being the only class that's over the 50% mark for profitability. Low-budget films are not considerably lower though, with only a 4% difference. Mid-budget films are yet again much lower than the other two classes. As this chart is counting the number of films that turned a profit, rather than the severity of that profit or loss, it stands to reason that a high-budget film is more likely to turn a profit than a low-budget film, though individual profits may be less sizable when in comparison with its budget. One factor that these two charts don't take into account however, is whether the film is a part of an established franchise.

Franchise ROI by Budget Class

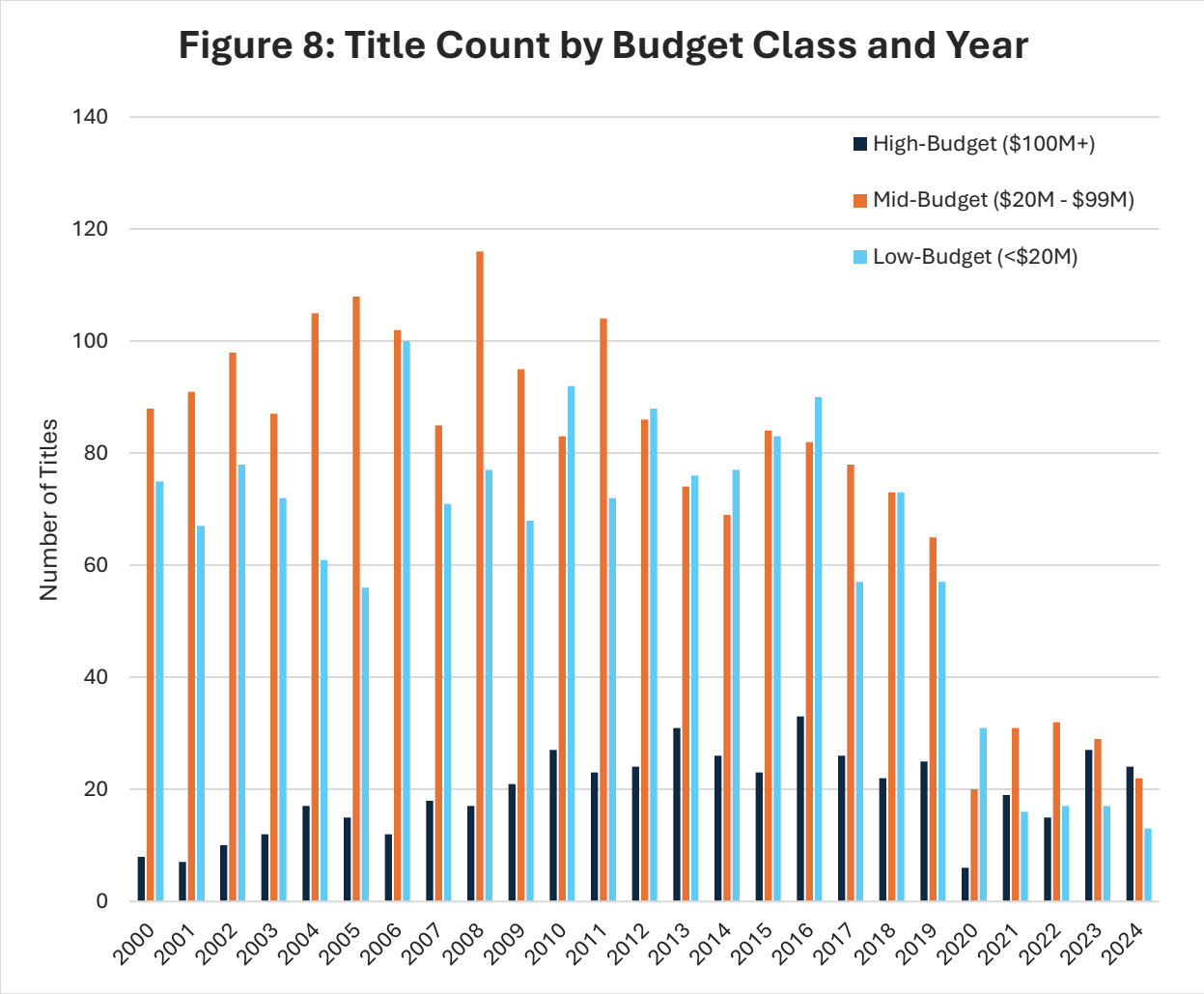
To assess the value of a film being part of a franchise, I split out the ROI by Budget analysis into Franchise and Standalone entries.



Franchise ROI for each budget category shows higher return than its corresponding standalone ROI. This conforms with the current landscape of the film industry, where a high number of franchise films are produced and hold a marketing advantage, as audiences have a pre-existing knowledge and affection for the material compared to standalone films. However, there are a few noteworthy observations with the low-budget ROI values. First, the standalone ROI for low-budget film is not only much higher than the other standalone ROI values, but also than the franchise ROI values for those categories. This further cements how much higher the ROI is for the low-budget category. Second, the low-budget franchise ROI is considerably higher than any other subcategory, despite franchise filmmaking being primarily associated with big budget superhero films. Given the top genres shown in Figure 3, we can infer that the low-budget franchise films are primarily horror films, which consistently turn a profit and are often made for a low-budget, notably including the massive success of the *Paranormal Activity* franchise and the films coming from Blumhouse Productions. It is worth noting that the first entry in a franchise, such as *Saw* (2004), is labeled in our dataset as a franchise film, despite the first film in a series not carrying the built-in benefits of being a part of a franchise. Having confirmed the value of a franchise, I then decided to examine another commonly held industry opinion: that fewer films are being made today than were being made 20 years ago.

Title Count by Budget Class and Year

Lastly, I generated a Pivot Chart showing the number of films being made per year, split by budget category.



While at the start of the millennium the vast majority of films were either low-budget or mid-budget films, more and more high-budget films were made each year, ultimately becoming the majority budget class in 2024. However, as the number of high-budget films produced increases, not only do the number of low-budget and mid-budget films decrease, but the overall number of films being made also has steadily decreased each year. While this trend seems to have been exacerbated by the 2020 pandemic, the trend was already present beforehand. The data indicates that as budgets increase, and more and more high-budget films get made, the total number of films being produced decreases, placing greater pressure on every released title to turn a healthy profit. This trend results in a greater risk concentration across the whole industry, and certainly one of the main contributors to the current instability facing the film industry. Producing a great number of films, and with a substantial number of low-budget films, would greatly alleviate this heightened risk.

Given the observed trends within these charts, I am now prepared to synthesize these findings into a conclusion and recommendation on which category of films are the safest investments.

5. CONCLUSIONS AND RECOMMENDATIONS

Given our findings, studios should produce more low-budget films. While high-budget films may be slightly more likely to turn a profit, the potential return relative to risk is substantially higher with low-budget films. Hypothetically, if you allocated $100 million towards either one big-budget film or ten to fifteen low-budget films, you would likely see a higher total return on investment by financing the group of low-budget films, even if the individual success rate might be higher on the big-budget film. Using our average ROI values by budget class, a $100 million film will on average return $154 million, while a group of low-budget films totaling $100 million will on average return, as a group, $318 million, even if not every film in that group is a hit.

Low-budget films that either start or continue a franchise are more likely to succeed. However, it is worth noting that while franchise low-budget films show higher ROI than standalone low-budget films, low-budget standalone films still significantly outperform mid and high-budget franchises, meaning low-budget films are less dependent on IP than the rest of Hollywood.

Genre has an impact as well. Low-budget films that also fall into one of the top genres, namely musical, adventure, action, and horror, are also more likely to succeed. Of those genres, horror is currently the genre being focused on within the low-budget filmmaking world and remains the safest bet, though the relative absence of low-budget musical, adventure, and action films could present an untapped opportunity to reach audiences underserved within this budget class.

It also should be noted that this analysis exclusively focuses on theatrical releases and budgets; films released on streaming, as well as the diminishing theatrical windows are beyond the scope of this study. As not every streaming service makes their viewership data public, it would be difficult to statistically ascertain concrete conclusions industry-wide on this front.

With the film industry struggling to recapture audiences, I believe the path forward is financing more films with modest budgets, rather than committing to big, potentially risky tentpole releases as the primary release type. Increasing the number of films being produced is crucial to alleviating the current high risk concentration felt across the film industry, and the only way to do that reliably is by greenlighting more low-budget films within a variety of genres, including both genres that currently perform well in that budget class, like horror, as well as popular genres that are currently being left solely to bigger budgeted productions, including action, adventure, and musical. This shift towards low-budget and high-volume financing could simultaneously reshape industry risk tolerance and foster creative experimentation, opening doors for new, exciting voices and talents. It’s the rare opportunity where the data points towards a solution that benefits the artistic and financial alike.

6. APPENDIX

- Kaggle Dataset: **Movies Metrics, Features and Statistics** by Michael Matta
- SQL Files: **GitHub SQL Folder**
- Excel Spreadsheet: **GitHub Excel Folder**